

Week 7 Assignment Report

Delta Lake - Incremental Load and SCD using MERGE

Objective

The goal of this assignment was to understand how Delta Lake handles incremental data updates using the MERGE operation. Starting with a customer dataset, the data was cleaned and then updated with a second dataset containing new and modified records.

Step 1 - Load Data

The customer master CSV file was loaded into Spark and written as a Delta table. This became the main table used throughout the assignment.

Step 2 - Clean the Data

Basic cleaning was performed before any updates. Duplicate customer IDs and the record with a missing email were removed so the master table contained only valid data.

Step 3 - Incremental Updates

The incremental CSV was then loaded. Using Delta Lake MERGE, existing customers were updated while new customers were inserted. The same dataset was also used to demonstrate SCD Type 2, where changed records were expired and new versions were added to preserve history.

Step 4 - Validation

After the merge, row counts and duplicate checks were performed to confirm that the pipeline produced the expected results.

Step 5 - Final Result

The final SCD Type 1 and SCD Type 2 tables were displayed to verify that updates and inserts had been applied correctly.

Summary

This assignment covered loading data, cleaning it, processing incremental changes, validating the output, and understanding the difference between SCD Type 1 and SCD Type 2 using Delta Lake MERGE.

Conclusion

Working on this assignment gave me a practical understanding of Delta Lake's MERGE functionality. While SCD Type 1 is useful for keeping only the latest information, SCD Type 2 is helpful when historical records need to be preserved.